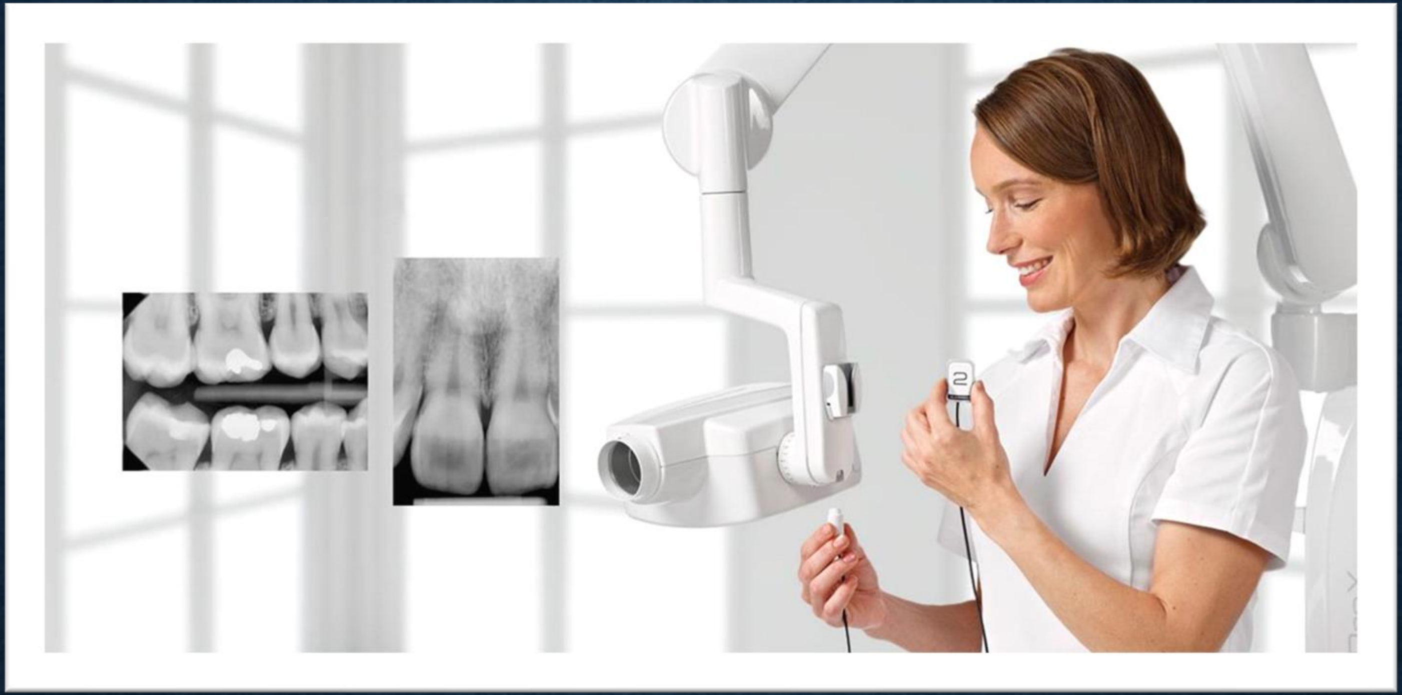


INTRAORAL PROJECTIONS

Dr.Manila

RADIOGRAPHIC IMAGES

- The radiographic images of the teeth, jaws and skull are divided into two main groups:
- Intraoral radiograph:
  - The film/image receptor is places inside the patient's mouth
- Extraoral radiograph:
  - The film/image receptor is places outside the patient's mouth

Intraoral-PA

Intraoral-BW

Extraoral-Ceph

Extraoral-Pano

EXTRAORAL RADIOGRAPH

- Panoramic radiographs:
  - Overall evaluation of dentition
  - Examine for pathology, such as cyst, tumors
  - Gross evaluation of temporomandibular joint
  - Evaluation of position of impacted teeth
  - Evaluation of eruption of permanent dentition
  - Trauma

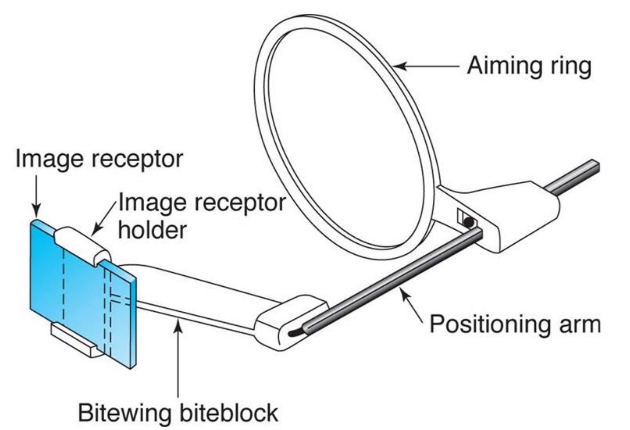
- Lateral skull radiographs:
  - Treatment planning (orthodontic treatment)
  - Evaluation of the skull, paranasal sinuses, airway, cervical vertebrae

INTRAORAL RADIOGRAPH-CRITERIA OF QUALITY

- Radiographs should record the complete areas of interest on the image
- Radiographs should have the least possible amount of distortion
- Images should have optimal density and contrast to facilitate interpretation

INTRAORAL RADIOGRAPH

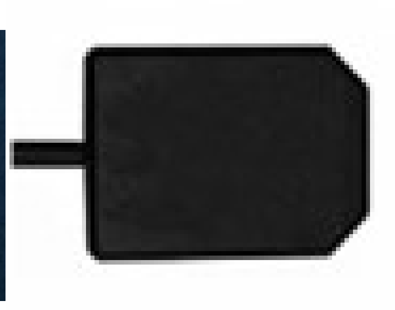
- Intraoral projections
  - Paralleling technique
  - Bisecting angle technique
- Image receptors
  - medium used to capture an image
    - CCD, CMOS, film etc
- Receptor holing instruments
  - positioning of the receptor
  - stabilize the receptor to a bite block
  - external guiding ring



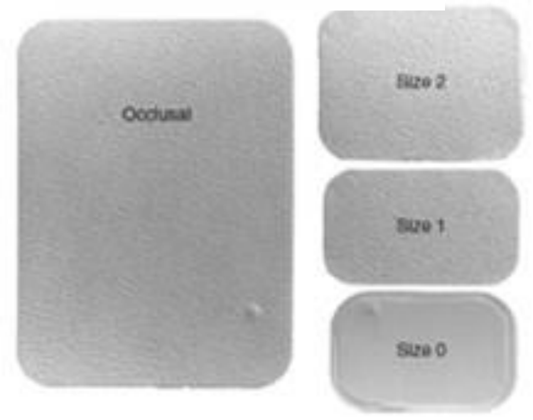
MATERIAL-INTRAORAL RECEPTOR

- Intraoral image receptor:
- Conventional receptor
- Digital receptor
  - CCD, COMS, PSP
- Films- various sizes-
  - Size 2: Adult posterior
  - Size 1: adult anterior
  - Size 0: Child size

Rigid CCD sensor

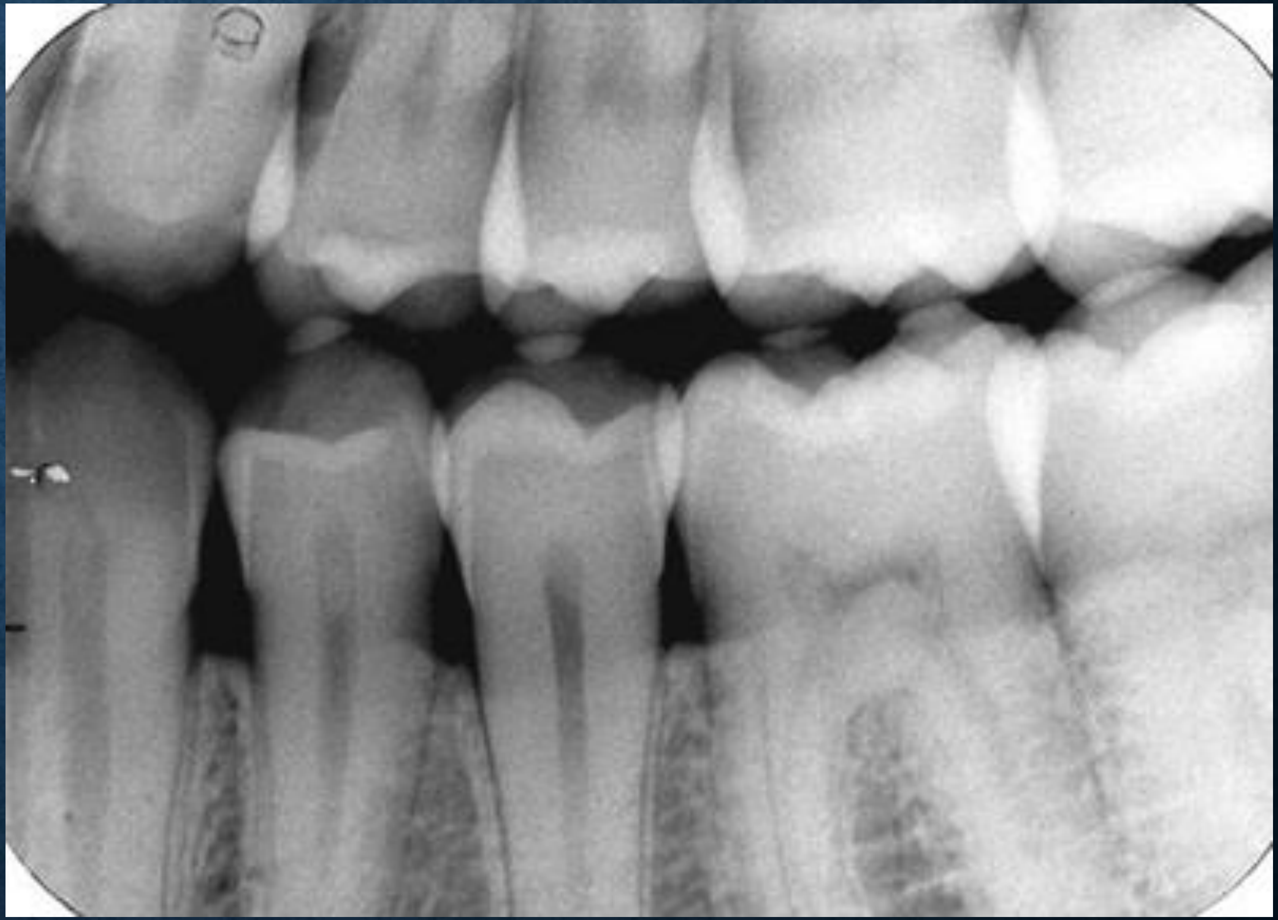


Dental x-ray films



INTRAORAL RADIOGRAPH

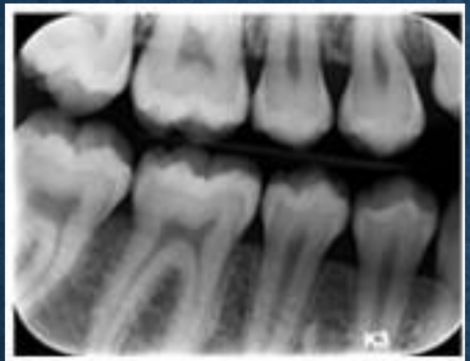
- Receptor placement
  - Parallel to the teeth & deep into lingual vestibule or palatal vault
  - Anatomic variations
    - shallow palate/FOM, tori
- Angulation of the tube head
  - Opening of the cylinder must be parallel to the plane of aiming ring
  - Horizontal direction of the beam influences the degree of overlapping



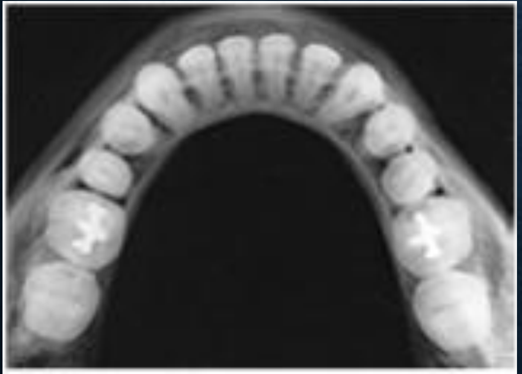
Horizontal overlapping of crowns results from misdirection of the central ray in the horizontal plane

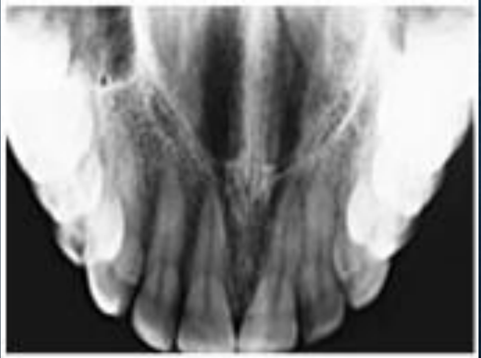
INTRAORAL RADIOGRAPH

Periapical radiograph

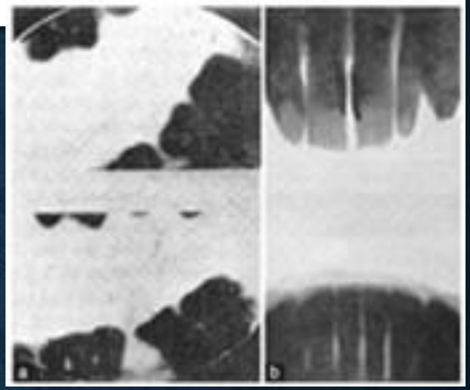


Bitewing radiograph





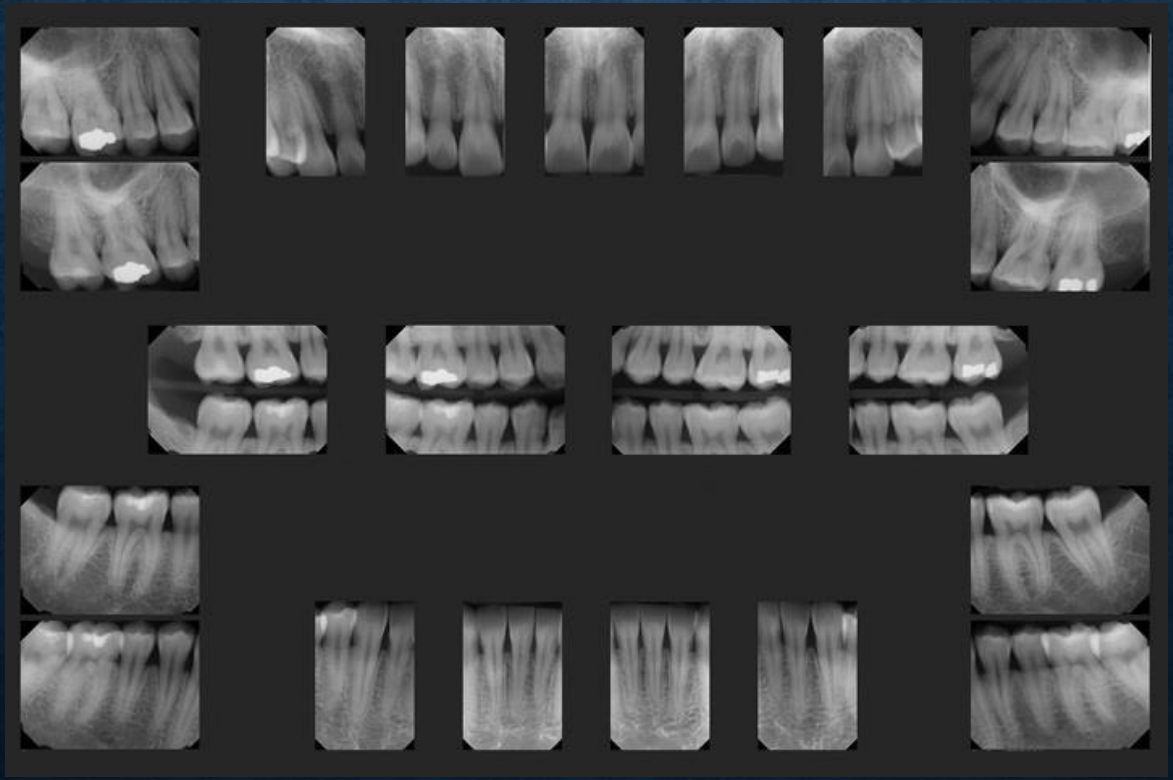
Occlusal radiograph



The first dental radiograms obtained in Germany by Otto Walkhoff (a) and Wilhelm Koenig (b)

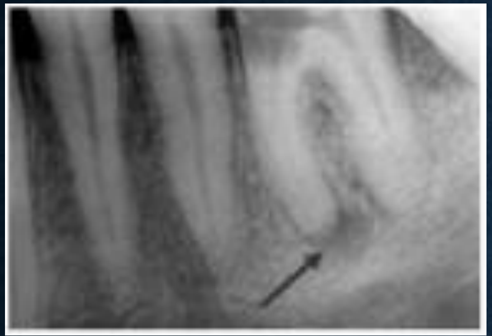
FULL MOUTH SET OF RADIOGRAPHS

- The full mouth adult survey consists of PA and BW images
- Typical full mouth series consists of 21 images: 17 PA and 4 BW



INTRAORAL RADIOGRAPH-PERIAPICAL RADIOGRAPH

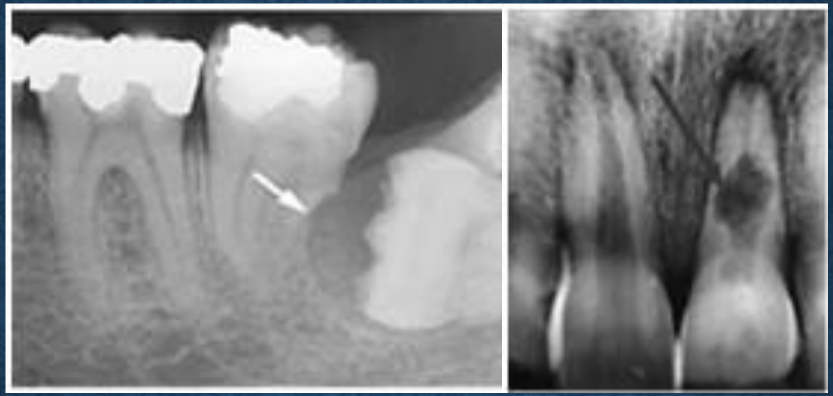
- Shows teeth and the tissue around the apices
- The main clinical indications include:
  - Detection of apical infection/inflammation: apical cyst or lesion
  - Assess periodontal bone loss
  - Evaluate unerupted and impacted teeth
  - Evaluate root morphology before extractions
  - Evaluate external and internal root resorptions
  - Root canal treatment
  - Assess implant osseointegration and peri-implant bone loss



Apical infection/apical cyst

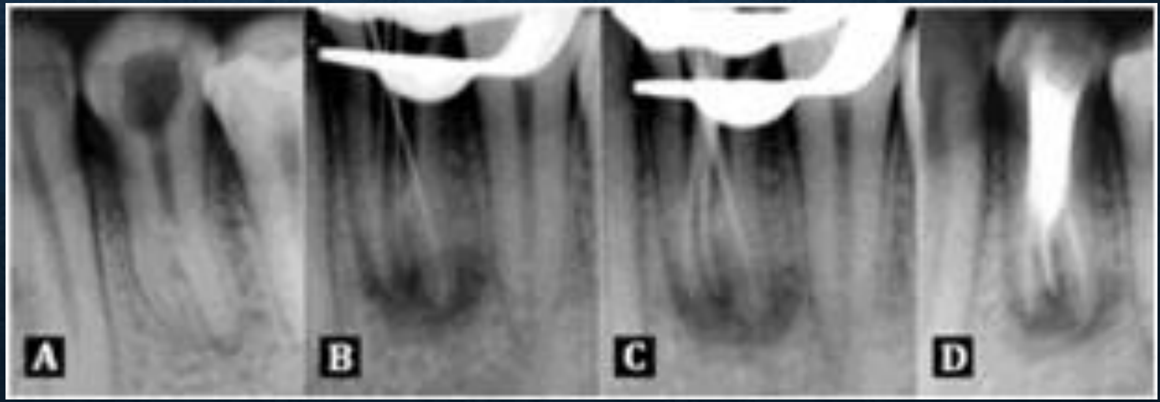
Root morphology/Dilacerated roots

Alveolar bone loss and cyst



External and Internal root resorption

Implant evaluation/peri-implant bone loss

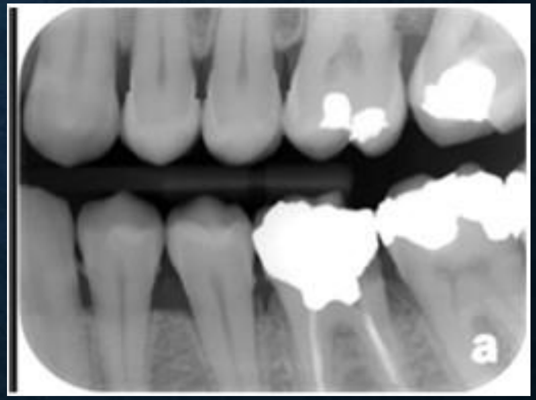


Root canal treatment

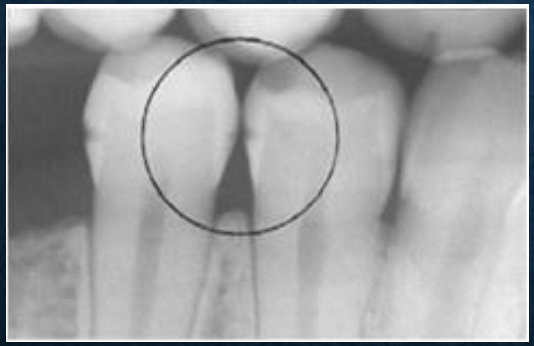
Impacted third molar

INTRAORAL RADIOGRAPHS-BITEWING RADIOGRAPHS

- Bitewing radiographs show only crown of teeth and adjacent alveolar crest
- The main clinical indications:
  - Detecting interproximal caries
  - Detecting interproximal calculus
  - Alveolar bone crest



Alveolar bone crest



Interproximal caries

Calculus

IDEAL IMAGE

- Radiographs should record the complete area of interest on the image
  - PA images: Shows full length of the roots and at least 2 mm of periapical bone
    - area of the entire abnormality plus some surrounding normal bone
  - BW images: Should show each posterior proximal surface at least once
    - Overlap of adjacent proximal tooth surfaces should be less than one third of the enamel thickness

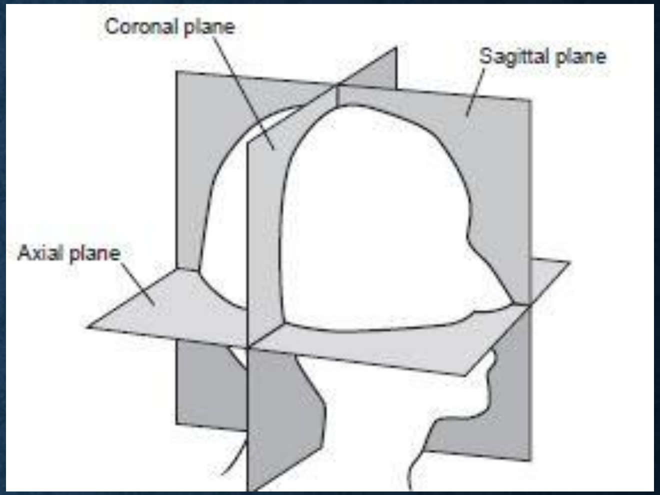
Incorrect film placement

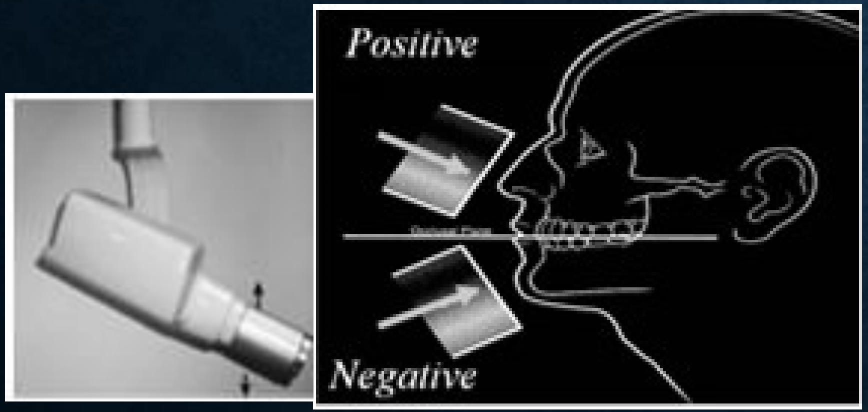
Short roots

Long roots

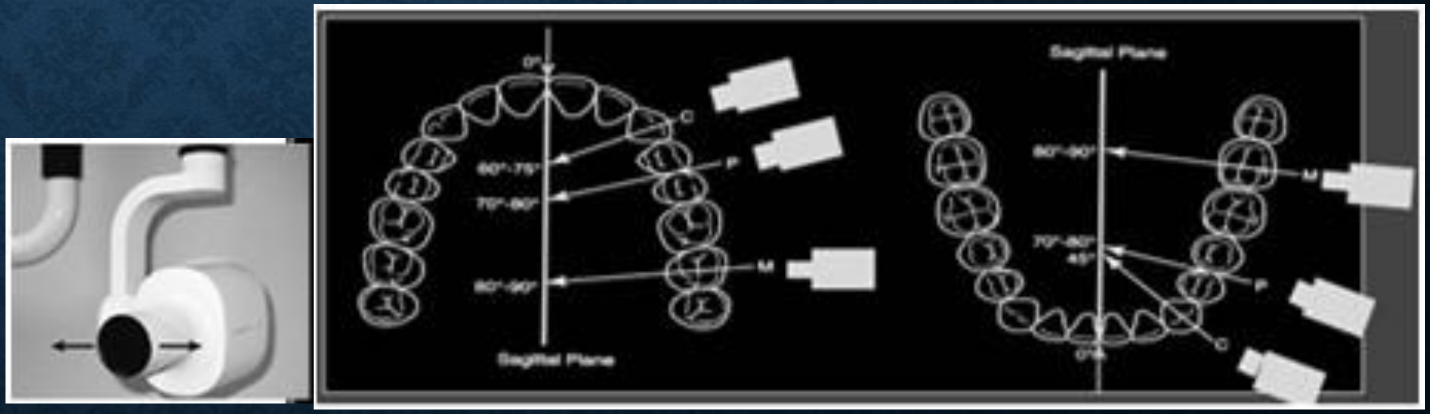
- Radiographs should have the least possible amount of distortion
    - Most distortion is caused by:
    - Inappropriate positioning of the receptor
    - Improper angulation of the x-ray beam

- Anatomic planes of the head:
    - Vertical (sagittal) plane
    - Horizontal (axial) plane
- X-ray beam angulation
    - Vertical angulation of the x-ray beam
    - Horizontal angulation of the x-ray beam





Vertical angulation

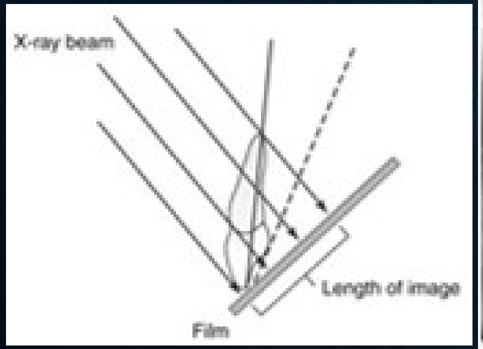


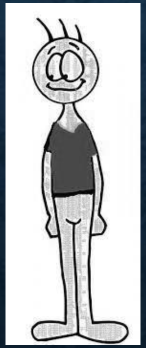
Horizontal angulation

Improper vertical angulation

Foreshortening

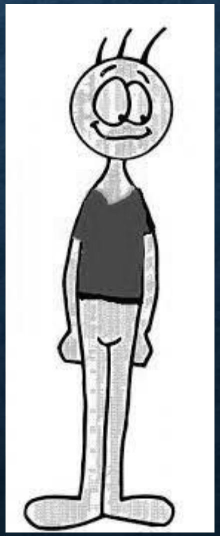
Excessive vertical angulation

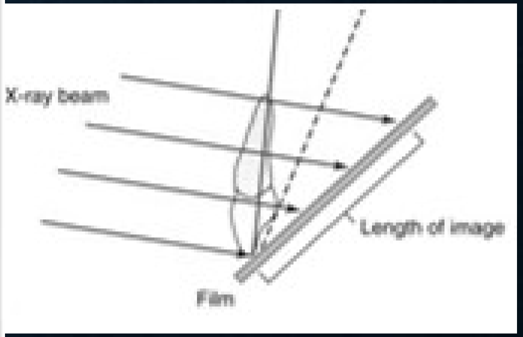




Elongation

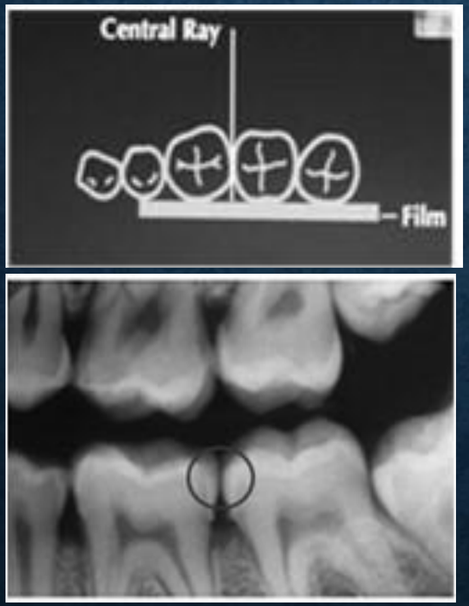
Insufficient vertical angulation





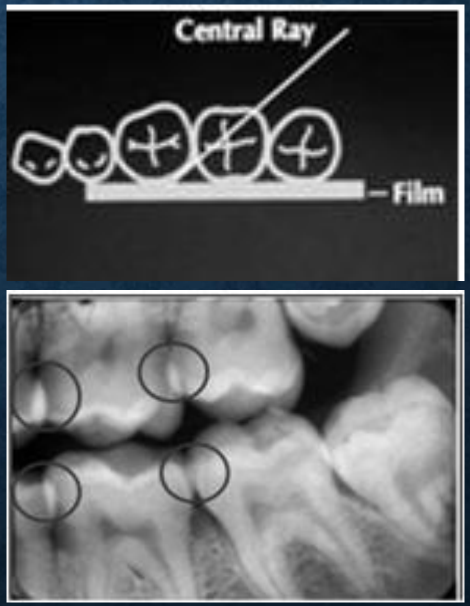
IMPROPER HORIZONTAL ANGULATION

- In the horizontal plane, the central ray should be aimed through the interproximal contact areas
- The result of improper horizontal angulation is proximal overlap



Proper horizontal angulation

Contacts open

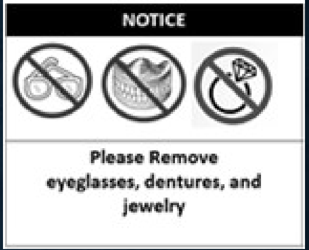


Improper horizontal angulation

Contacts overlapped

HOW TO PREPARE?

- Preparing patient
  - Review PMH, PDH, previous radiographs & physical exam
  - Positioning the patient
    - Maxillary projections: upright in the dental chair with the sagittal plane vertical and the occlusal plane horizontal
    - Mandibular projections: upright in the dental chair with the sagittal plane vertical. The head is tilted back slightly to compensate for the changed occlusal plane when the mouth is opened
  - Remove eyeglasses, dentures & jewelry
  - Place lead apron/thyroid collar
- Adjust the x-ray exposure parameters
  - kVP, mA and exposure time
  - Generally, only the exposure time is adjusted for the various anatomic locations

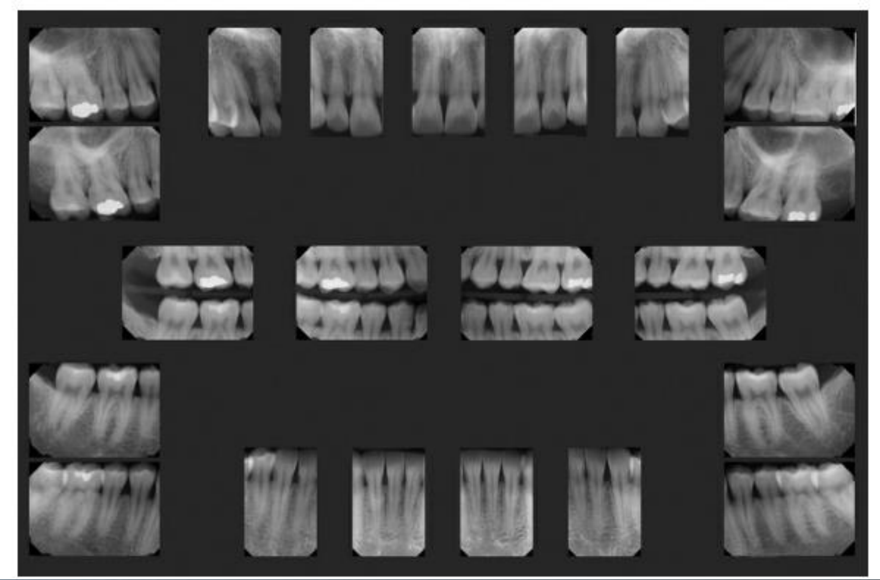


HOW TO PREPARE?

- Receptor placement
  - Behind the area of interest, with the apical end against the mucosa on the lingual or palatal surface
  - The occlusal or incisal edge is oriented
  - Against the teeth with an edge of the receptor extending just beyond the teeth
- Angulation of the tube head
  - Horizontal angulation
  - Vertical angulation

FULL-MOUTH RADIOGRAPHIC SERIES

- A typical full-mouth radiographic series consists of 21 images
- start with the anterior views because they cause less discomfort for the patient.



PROJECTIONS FOR A TYPICAL FULL-MOUTH RADIOGRAPHIC SERIES

Anterior Periapical (Use No. 1 Receptor)

- Maxillary central incisors: one projection
- Maxillary lateral incisors: two projections
- Maxillary canines: two projections
- Mandibular central and lateral incisors: two projections
- Mandibular canines: two projection

PROJECTIONS FOR A TYPICAL FULL-MOUTH RADIOGRAPHIC SERIES

Posterior Periapical (Use No. 2 Receptor)

- Maxillary premolars: two projections
- Maxillary molars: two projections
- Maxillary distomolar (as needed): two projections
- Mandibular premolars: two projections
- Mandibular molars: two projections
- Mandibular distomolar (as needed): two projections

OCCLUSAL RADIOGRAPHY

Displays a relatively large segment of a dental arch

Different types of occlusal radiographs:

- Cross-sectional maxillary occlusal projection
- Anterior Mandibular projection
- Cross-sectional Mandibular projection

Remember ...

There are several other types of occlusal radiographs!

DIAGNOSTIC OBJECTIVES OF OCCLUSAL RADIOGRAPHY

- To locate supernumerary, unerupted, and impacted teeth
- To localize foreign bodies in the jaws and floor of the mouth
- To identify and determine the full extent of disease (e.g., cysts, osteomyelitis, malignancies) in the jaws, palate, and floor of the mouth
- To evaluate and monitor changes in the mid-palatal suture during orthodontic palatal expansion. To detect and locate sialoliths in the ducts of sublingual and submandibular glands

OCCLUSAL RADIOGRAPHY

- To evaluate the integrity of the anterior, medial, and lateral outlines of the maxillary sinus
- To aid in the examination of patients with trismus, who can open their mouths only a few millimeters
- To obtain information about the location, nature, extent, and displacement of fractures of the mandible and maxilla

BASIC PRINCIPLES OF OCCLUSAL RADIOGRAPHY

- The occlusal film is positioned with the front side facing the arch being exposed.
- The film is placed between the occlusal surfaces of the teeth.
- The film is stabilized when the patient gently bites on the surface of the film.
- Occlusal receptors are made of film or storage phosphor plates.
- CCD or CMOS sensors of this size are not manufactured

IMAGING OF CHILDREN

- Greater sensitivity to radiation
- Make minimal number of radiographs
- Bitewing examinations -not required when the contacts are open & can be visually examined
- PA-survey -recommended early in the mixed dentition stage
- Use of fast receptor, proper processing, beam-limiting devices, and protective aprons and thyroid shields.

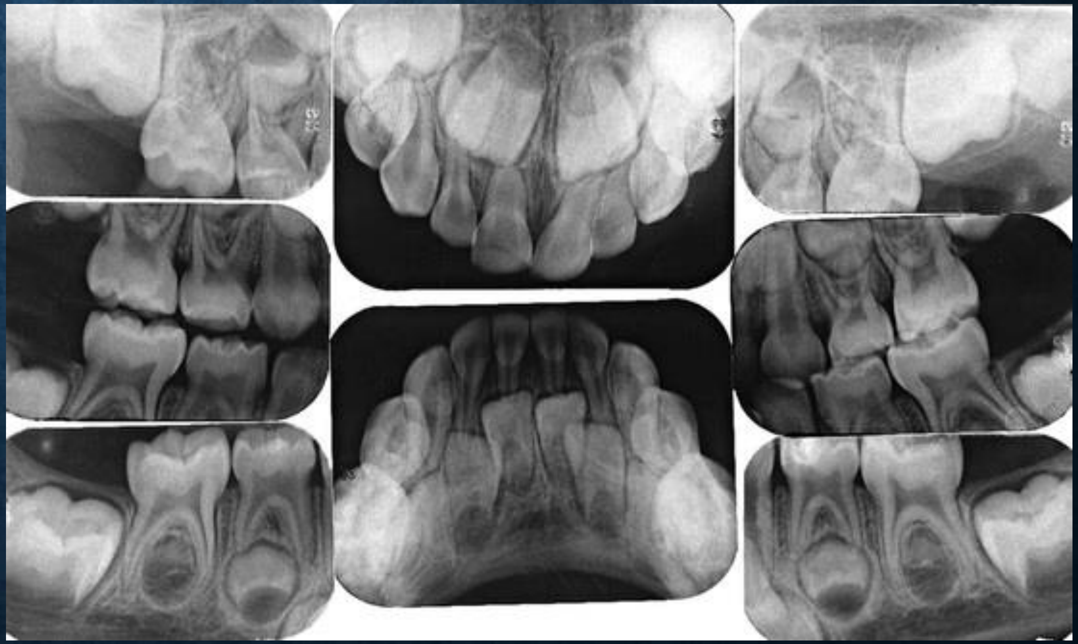
IMAGING OF CHILDREN

- Primary dentition (3-6 yrs)
  - Combinations of projections can be used,
  - No. 0 & No. 2 receptors used
  - 2 anterior occlusal, 2 post BW, 4 post P
  - Maxillary and interproximal projections: child is seated upright with the sagittal plane perpendicular to and the occlusal plane parallel with the floor
  - Mandibular projections: child is seated upright with the sagittal plane perpendicular
  - Tragus corner of the mouth line is oriented parallel to the floor.

RADIOGRAPHIC EXAMINATION OF PRIMARY DENTITION

- Two- anterior occlusal views
- Four -posterior periapical views
- Two- bitewing views

- Preferable to make a BW and panoramic examination followed by selected periapical views as indicated



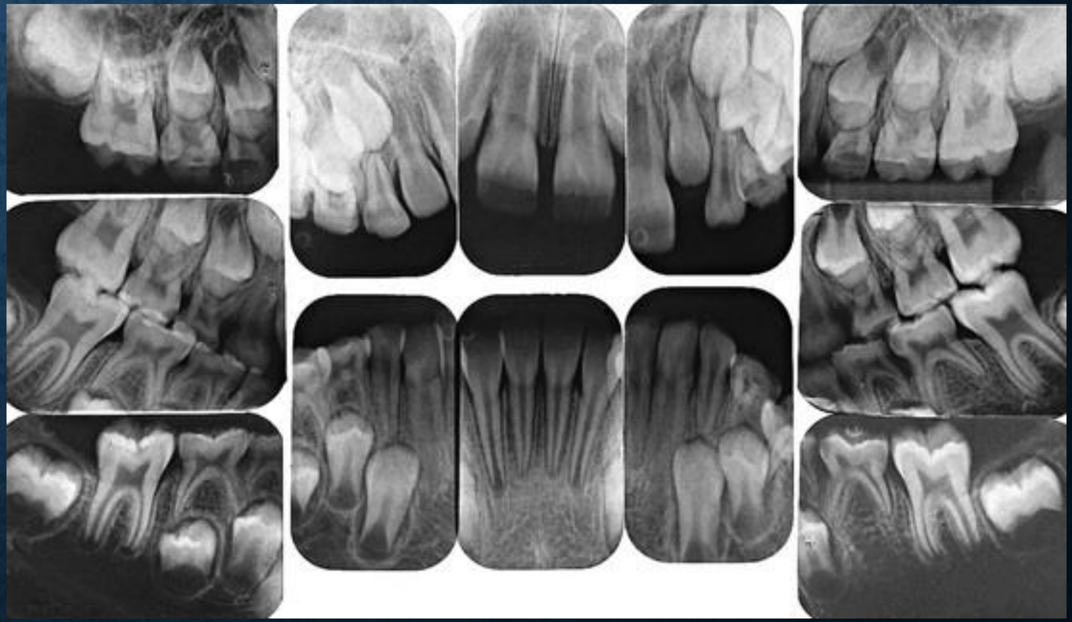
IMAGING OF CHILDREN

- Mixed Dentition (7 to 12 Years)
- two incisor PA, four canine PA, four posterior PA, and two or four posterior BW views
- No.1 & No.2 receptors are used
- XCP instruments are used for larger children
- Bisecting angle bite blocks may be more comfortable for smaller individuals.

RADIOGRAPHIC EXAMINATION OF MIXED DENTITION

- Two incisor views,
- Four canine views
- Four posterior views
- Two bitewing views.

- Often it is preferable to make a bitewing and panoramic examination followed by selected periapical views as indicated.



MOBILE INTRAORAL RADIOGRAPHY

- Remote sites such as nursing homes, hospitals, or at disaster scenes-portable machine is highly advantageous
- Provides rapid, self-contained imaging capability
- Approved by the U.S. Food and Drug Administration (FDA)
- Nomad handheld x-ray unit

NOMAD

- Unit can be held stable and produce clinically acceptable images.
- Uses a high-frequency constant-potential x-ray generator (60 kv constant potential) and has a short focal spot-to-skin distance (20 cm)—both allow for short exposure times compared with conventional units.
- Small focus spot (0.4 mm).
- Operator dose is mitigated using internal shielding materials in the unit to reduce leakage exposure and a shield on the aiming cylinder to minimize backscatter from the patient.
- Several states in the united states have additional requirements related to training and radiation dose documentation when using these handheld units.

SPECIAL CONSIDERATIONS –INTRAORAL IMAGING

- Radiographic procedures may need modifications for patients who have unusual difficulties or special needs.
- depend on the patient's physical and emotional characteristics.
- Infection -panoramic or occlusal techniques
- Trauma -panoramic or other extraoral views or CT
- Patients With mental or physical disabilities
- Gag reflex-relax and reassure patient
- Pregnancy

SPECIAL CONSIDERATIONS –INTRAORAL IMAGING

- Imaging for Endodontics
  - The tooth being treated must be centered in the image.
  - The receptor must be positioned as far from the tooth and apex as the region permits to ensure that the apex of the tooth and some periapical bone are apparent on the radiograph.
  - EndoRay receptor holder used for endodontic images. This device provides room for files to be in place while making the image.
  - Fit over files, clamps, and the rubber dam without touching the subject tooth

SPECIAL CONSIDERATIONS –INTRAORAL IMAGING

- Edentulous patients
  - Look for- Roots, residual infection, impacted teeth, cysts, or other pathologic entities
  - Panoramic examination –Most convenient
  - If not available- 14 intraoral views provides an excellent survey
  - Central incisors (midline): one projection
  - Lateral canine: two projections
  - Premolar: two projections
  - Molar: two projections